**9** **(a)** Fig. 9.1 shows a magnet on the end of a spring and a coil of wire connected to a sensitive centre-zero galvanometer. The magnet can move freely through the coil.

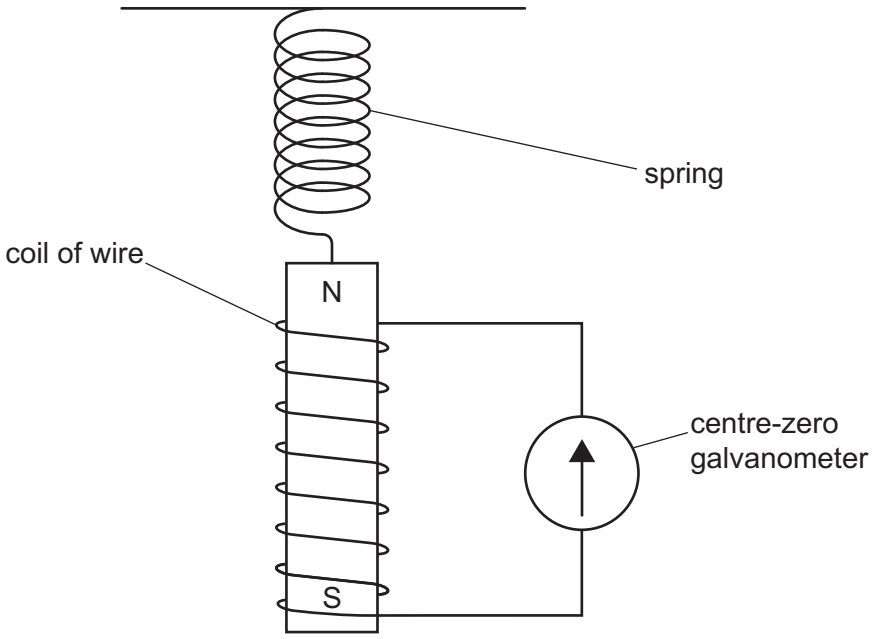



**Fig. 9.1**

**(i)** The magnet is pulled down and released.

Describe and explain what happens to the needle of the sensitive galvanometer.

.....

.....

.....

..... [4]

**(ii)** The magnet is replaced with a stronger magnet.

State the effect of using a stronger magnet on what happens to the needle of the galvanometer.

..... [1]

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**(b)** A step-up transformer is used to step up the output voltage of a power station from 25 000 V to 400 000 V for transmission along power lines.

The number of turns on the secondary coil is 36 000.

Calculate the number of turns on the primary coil.

number of turns = ..... [2]

[Total: 7]